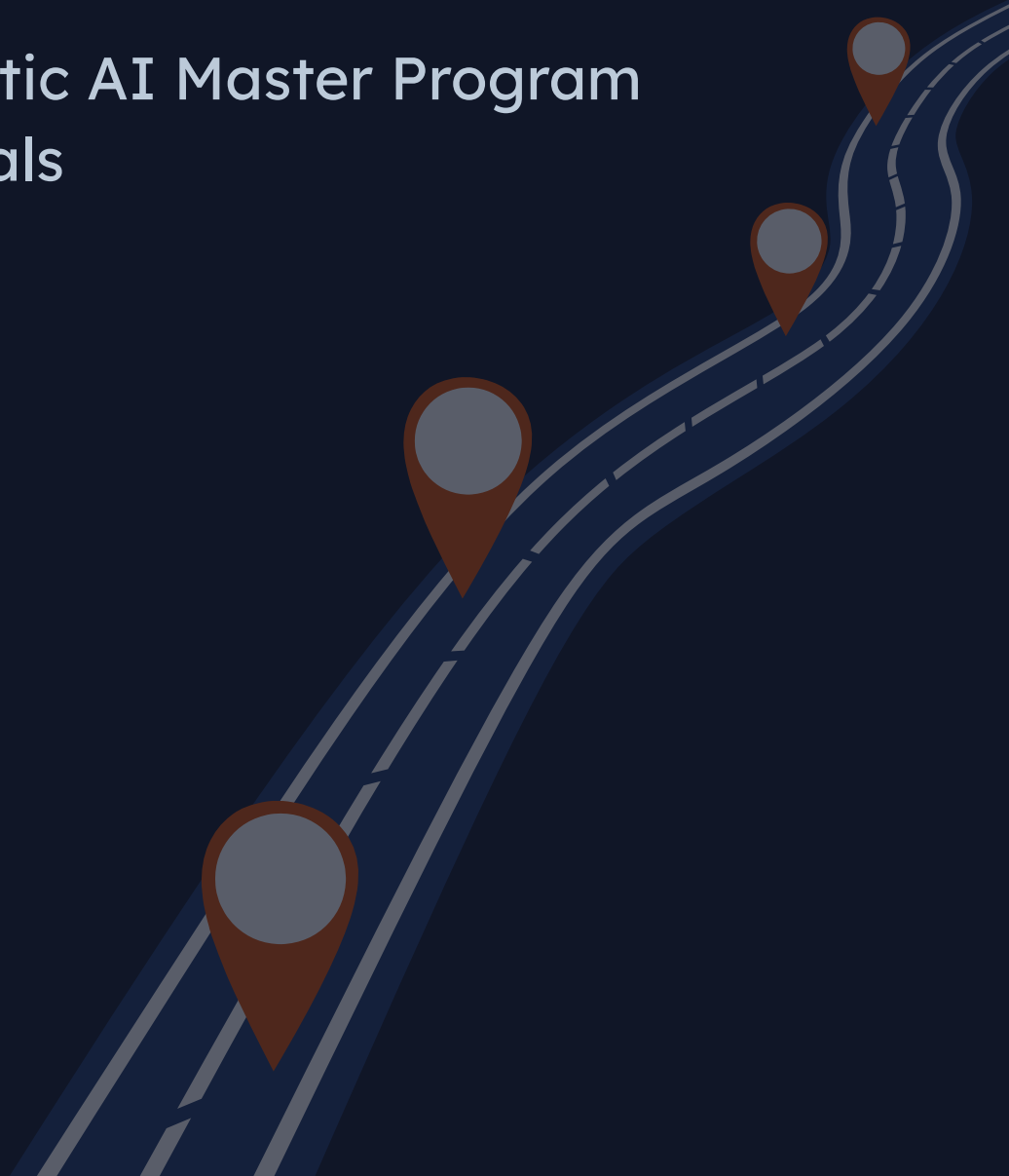


9 Months Program

Program Roadmap

GenAI & Agentic AI Master Program
for Professionals



Program Roadmap

FOR GENAI AND AGENTIC AI MASTER PROGRAM

DAY 0

DURATION: 3 DAYS

INDUCTION + ORIENTATION

Understand the GenAI & Agentic AI landscape, tools, expectations, and how this program aligns with your career goals.

MODULE 1

DURATION: 1 Months

PYTHON FOR GENAI

Build a strong Python foundation tailored for developing, integrating, and controlling GenAI workflows.

MODULE 2

DURATION: 1 Months

FOUNDATION FOR GENAI

Learn core ML, deep learning, and NLP concepts that power modern large language models.

MODULE 3

DURATION: 1.5 Months

ADVANCED GENERATIVE AI

Design, build, and optimize LLM-based applications using prompts, embeddings, RAG, and fine-tuning.

MODULE 4

DURATION: 1.5 Month

AGENTIC AI & AUTOMATION

Create autonomous AI agents that plan, reason, collaborate, and execute tasks across systems.

MODULE 5

DURATION: 1 Months

LLMOP's & Deployment

Design, build, and optimize LLM-based applications using prompts, embeddings, RAG, and fine-tuning.

Module 6

DURATION: 2 Weeks

Ethics, Guardrails & Responsible AI

Learn how to design, deploy, and govern GenAI and Agentic AI systems responsibly using safety, compliance, and human-in-the-loop guardrails.

Domain Specialization & Electives

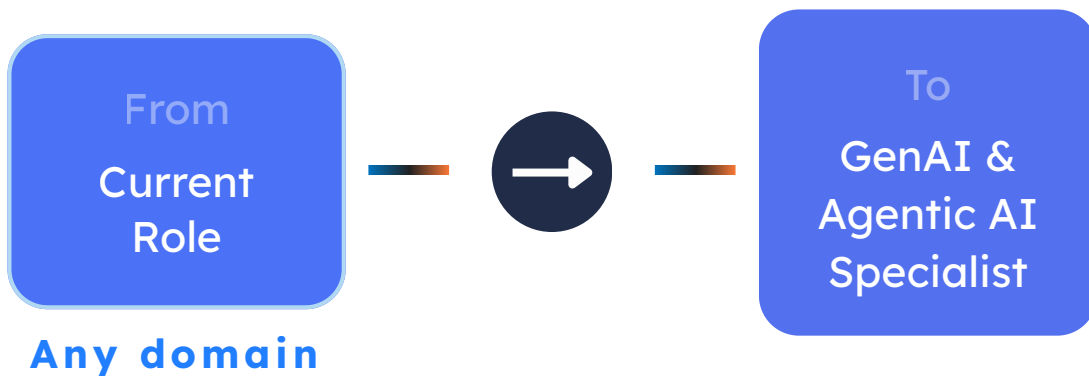
DURATION: 2 Months

Work on AI Startup Projects

Work on real AI startup projects to build deployable systems and earn industry-recognized certifications.

Domain Specialisation

(2 Months)



DOMAIN SPECIALISATIONS (CHOOSE ANY 2)

Elective A

4 Project

GenAI and Agentic AI for BFSI

AI solutions for fraud detection, credit risk modeling, customer intelligence, underwriting automation, and financial copilots.

Elective B

4 Project

GenAI for Retail & ecommerce

Build AI systems for demand forecasting, personalized recommendations, inventory optimization, and customer experience automation.

Elective C

4 Project

GenAI for Manufacturing & Supplychain

Apply GenAI for predictive maintenance, quality inspection, supply chain intelligence, and factory optimization.

Elective D

4 Project

GenAI for Healthcare & Pharma

Develop AI use cases for clinical decision support, medical document analysis, diagnostics assistance, and patient data intelligence.

Elective E

4 Project

GenAI for Cloud & Devops

Design autonomous agents, AI-driven CI/CD, monitoring agents, RAG pipelines, and production-grade MLOps systems.



Elective F

4 Project

GenAI for Managers & Leaders

Learn Project Management using AI, Gen AI for Digital Business Transformation , AI governance, agent-based decision systems.

Elective G

4 Project

GenAI for Software Developers

Build LLM-powered applications, APIs, copilots, autonomous agents, and GenAI features using modern frameworks.

Elective H

4 Project

GenAI for QA & Test Engineers

Build AI systems for test case generation, defect triage, coverage analysis, and automated quality intelligence.



Saurabh Kumar

From
Professor



To
**Data
Scientist**



Placed at
Teleperformance

**135%
Salary
Hike**

9 Months Program

Program Curriculum

GenAI & Agentic AI Master Program
for Professionals

140+ Hours of
Industrial Projects

24+ Hours of
Capstone Projects

Unlimited Doubt
Clearing Sessions

Induction + Orientation

Duration: 3 Days

Module 0

- Overview Of AI, Generative AI, And Agentic AI.
- Career Paths And Role Mapping For AI Professionals.
- LinkedIn Profile Optimization For AI & Tech Hiring.
- Using **Python Colab** For Hands-On Learning.
- Learning Workflow, Tools, And Program Onboarding.

Python for GenAI

Duration: 1.5 Months

Module 1

Topic 1: Core Python

- **Variables**
 - Store values in various data types
 - Dynamic assign / reassign variables
 - Python naming conventions for maintainable code
- **Data types**
 - Numeric types: **int**, **float**, **complex**.
 - Text handling using str and string methods.
 - Collections: **list**, **tuple**, **set**, **dict**.
 - Mutable vs immutable data behavior.
- **Loops**
 - Iteration using for and while loops.
 - Sequence traversal with **range()**, **enumerate()**, **zip()**.
 - Optimizing iterations for data processing tasks.

Topic 1: Core Python

- **Control Statements**

- Conditional logic using if, elif, else.
- **Comparison operators** (==, !=, <, >, <=, >=).
- Logical operators (and, or, not).
- Flow control using break, continue.
- Building **decision trees** for real-world logic handling.

- **Functions**

- Creating reusable logic blocks using def and return.
- Passing parameters using positional and keyword arguments.
- Anonymous function with **Lambda**

Project 1 Simple Calculator

1 hour

Create a console-based calculator that takes user input for two numbers and performs mathematical operations

Topic 2: Advanced Python

- **File Handling**

- Reading and writing files using open(), read(), write().
- Handling text, **CSV**, and **JSON** files for data pipelines.
- Working with file paths using os and pathlib.

- **Regex**

- Pattern matching using **re.search()**, **re.findall()**, **re.sub()**.
- Cleaning and extracting text data from logs, emails, and documents.
- Using regex tokens: \d, \w, +, *, ^, \$, groups.

- **Exceptional Handling**

- Custom exceptions handling using try, except, finally.
- Common exceptions.
- Preventing pipeline failures in production code.

Topic 2: Advanced Python

- **OOP's**

- Creating classes and objects using class and `__init__`.
- Instance variables vs class variables.
- Methods and object behavior modeling.
- Inheritance and method overriding.

Project 2 Banking System Simulation

BFSI

Simulate a simple banking system where users can create accounts, deposit, withdraw, and check balances.

Python

Data Structure

Function

Project 3 Movie Night Recommender

Entertainment

Recommend movies based on genre preferences using simple rule-based logic and conditional checks.

Data Types

Filtering Logic

User Input Handling

Project 4 Smart Fitness Habit Tracker

Healthcare

Track daily exercises, calories burned, and generate habit summaries using basic loops and functions.

Python

Functions

Conditional logic

Topic 3: Essential Libraries for GenAI

- **Numpy**
 - Creating arrays **using array, arange, linspace**.
 - **Array operations:** indexing, slicing, reshaping.
 - Vectorized computations for performance.
 - **Mathematical functions:** mean, sum, std, dot.
 - Broadcasting and handling multi-dimensional data.
- **Pandas**
 - Working with Series and DataFrame.
 - Reading data from **CSV, Excel, JSON**.
 - Data cleaning: missing values, duplicates, type casting.
 - Filtering, sorting, and conditional selections.
 - **Grouping** and **aggregation** using **groupby**.
 - Feature preparation for ML and GenAI pipelines.
- **Matplotlib**
 - Creating line, bar, scatter, histogram plots etc.
 - Customizing labels, titles, legends, and axes.
 - **Visualizing** trends and distributions in datasets.
- **Seaborn**
 - **Statistical plots:** boxplot, violinplot, countplot.
 - Relationship analysis using **pairplot** and **heatmap**.
 - Visualizing correlations and feature importance.
 - Styling plots for reports and dashboards.

Project 5 Retail Sales Data EDA

Retail

Analyze a retail sales dataset: clean data, plot trends (time series, bar charts), and derive insights.

Python

Pandas

Geopandas

Matplotlib

Topic 1: ML Fundamentals

- **Basic of ML**
 - **Supervised** vs **Unsupervised** learning paradigms.
 - Features, labels, training, validation, and test sets.
 - Bias-variance trade-off and model generalization.
 - **Overfitting** and **underfitting** with real examples.
 - End-to-end ML workflow used in industry projects.
- **Regression Analysis**
 - **Multiple Regression** with feature interactions.
 - Model evaluation using **RMSE, MAE, R² score**.

Project 6 Energy Consumption Forecasting

Energy

Predict household energy consumption using linear regression by analyzing temperature, appliances, and daily usage patterns.

Feature engineering

RMSE/R² evaluation

- **Classification Analysis**
 - **Logistic Regression** for binary classification.
 - Handling class imbalance using sampling techniques.
 - **Evaluation metrics: Precision, Recall, F1-Score, ROC-AUC.**

Project 7 Loan Approval Prediction

Finance

Build a classification model (e.g. logistic regression, random forest) to predict loan approvals from applicant data.

scikit-learn

Pandas

Topic 2: Deep Learning Fundamentals

- **Perceptron**
- **Backpropagation and Optimization**
- **Attention Mechanism**

Project 8 Smart Document Relevance Classifier

HR

Compute attention weights on simple sentences to understand token importance during context interpretation.

Numpy

Perceptron

Topic 3: NLP Fundamentals

- **Text Pre-Processing for Gen AI**
- **Text Embedding Techniques**
 - Bag-of-Words representation.
 - TF-IDF vectorization.
 - **Word2Vec: CBOW and Skip-Gram.**
- **Named Entity Recognition (NER)**
- **Topic Modeling**
 - Latent Dirichlet Allocation (LDA).
 - Latent Semantic Techniques (LSA/LST).
- **Part-of-Speech Tagging (POS)**
- **NLP Evaluation Metrics**

Topic 3: NLP Fundamentals

- **Transformer - BERT Model**

- **Transformer architecture:** Encoder-Decoder blocks.
- Multi-head self-attention mechanism.
- Positional encoding concepts.
- BERT pre-training objectives: **MLM** and **NSP**.
- Use cases of **BERT** in real NLP systems.

Project 9 Sentiment Analysis Using BERT

Marketing

Use pretrained BERT embeddings to convert sentences into vectors and classify sentiment using a simple classifier.

BERT

Manual tokenization

Dense classifier

Project 10 Legal Entity Extraction

Legal

Perform named entity recognition and classification in legal documents using spaCy, and visualize results

spaCy

Pandas

Matplotlib

Project 11 Customer Churn Prediction

Telecom

Predict churn using ensemble models + SHAP for explainability

scikit-learn

XGBoost

SHAP

Topic 1: Fundamentals of GenAI

- Difference Between Predictive AI And Generative AI.
- Generative Modeling Concepts And Probability Distributions.
- Pre-training vs Fine-tuning Paradigms.
- Foundation Models And Their Role In Modern AI Systems.

Topic 2: Use cases

- **Text Generation** For Chatbots And Assistants.
- Document Summarization And Report Automation.
- **Code Generation** And Developer Productivity Tools.
- Content Generation For Marketing And Media.
- Knowledge Assistants For Enterprises.
- **Search Augmentation** And **Question Answering Systems**.

Topic 3: Prompt Engineering

- **Zero-shot, One-shot, And Few-shot** Prompting.
- Instruction-based Prompt Design.
- Chain-of-Thought Prompting Techniques.
- Prompt Templates And Reusability.
- **Prompt Evaluation** And **Iterative Optimization**.

Topic 4: LLM

- **GPT Family** (GPT-3.5, GPT-4, GPT-4o).
- **LLaMA, Claude** and **Open-Source LLM** Ecosystem.
- **Tokenization** and **Context Windows**.

Topic 4: LLM

- **Inference Parameters: Temperature, Top-p, Max Tokens.**
- Hallucination And Limitations Of LLMs.
- Cost And Latency Considerations In Production.

Project 12 AI-Powered Sales Email Writer

Sales

Use an LLM to generate personalized sales outreach emails using customer behavior and CRM attributes.

OpenAI

Prompt Templates

Langchain

Topic 5: Hugging Face

- **Transformers Library** For NLP And GenAI.
- Pre-trained Models Hub And Model Cards.
- **Tokenizers** And **Pipeline** APIs.
- Fine-tuning Models Using **Hugging Face Trainer**.

Project 13 Product Review Sentiment Classifier

Retail

Build an NLP model to classify customer product reviews as positive or negative (using word embeddings and an LSTM or transformer).

NLTK/spaCy

Hugging Face

NLP Libraries

Topic 6: RAG (Retrieval Augmented Generation)

- **Vector DB**
 - Embedding Generation Using OpenAI And Hugging Face Models.
 - Vector Stores: **FAISS, Pinecone, ChromaDB, Weaviate, Qdrant**
 - Similarity Search And Distance Metrics.
 - **Chunking Strategies** For Optimal Retrieval.
- **Multimodal RAG**
 - Text + Image Retrieval Pipelines.
 - Multimodal Embeddings (**CLIP, Vision Transformers**).
 - Use Cases In Documents And Media Search.
- **Graph RAG**
 - Knowledge Graph Construction From Documents.
 - Relationship-based Retrieval **Over Vector Search**.
- **RAG Evaluation Metrics**
 - Context Precision And Recall.
 - Answer Faithfulness And Groundedness.
 - **Latency And Cost Evaluation**.
 - Human-in-the-loop Evaluation Methods.

Project 14 Custom Chatbot with Retrieval

Tech Support

Build a domain-specific chatbot: fine-tune an LLM on company FAQs and implement retrieval-augmented generation (Chroma or Elasticsearch) to answer user queries.

OpenAI API

Langchain

ChromaDB

Project 15 HR Policy RAG Query Bot

HR

Create a RAG-based chatbot that answers employee HR queries using internal company policies and guidelines.

Vector Store

Langchain

RAGAs Evaluation

Topic 7: Langchain / Llama Index

- **Prompt Templates And Output Parsers.**
- Chains And Sequential Workflows.
- Agents And Tool Calling.
- Integration With Vector Databases And APIs.
- Embed and store documents in vector DBs (**Pinecone, FAISS, Chroma**, etc.)
- Retrieve relevant chunks based on a user query
- Format the retrieved chunks as context for the LLM

Topic 8: Fine tuning

- **Full Fine-Tuning vs Parameter-Efficient Fine-Tuning (PEFT).**
- **LoRA And QLoRA** Techniques.
- Instruction Fine-Tuning.
- Dataset Preparation And Labeling Remotely.
- **Overfitting Risks And Mitigation Strategies.**

Topic 9: Fundamental Generative Models: GAN And VAE

- **Generative Adversarial** Networks Architecture.
- Variational Autoencoders And Latent Space Learning.
- Use Cases In **Image And Data Generation.**

Project 16 Synthetic Defect Image Generation

Manufacturing

Generate synthetic defect images to augment limited manufacturing datasets for improved model training performance.

GAN

Pytorch

TensorFlow

Topic 10: Stable Diffusion

- **Diffusion** Model Fundamentals.
- **Text-to-Image** Generation Pipelines.
- Prompt Control And **Image Conditioning**.
- Fine-Tuning Diffusion Models (**DreamBooth**, **LoRA**).

Project 17 Stable Diffusion Image Generator

Marketing

Use a Stable Diffusion model to generate creative images from text prompts; experiment with fine-tuning or style transfer.

CUDA GPU

Diffusers library

Topic 11: Flux Framework

- Image And Creative **Generation Workflows**.
- High-quality **Diffusion-based Outputs**.

Topic 12: Azure OpenAI

- **Azure OpenAI** Service Architecture.
- Model Deployment And Endpoint Management.
- **Enterprise Security** And Compliance.
- Integrating Azure OpenAI With Applications.

Project 18 Enterprise Document Assistant

Enterprise IT

Build a secure document Q&A system using Azure OpenAI embeddings and enterprise-grade retrieval pipelines.

Azure OpenAI

Python SDK

LangChain

Agentic AI & Automation

Duration: 1.5 Months

Module 4

Topic 1: Understanding Agentic AI and Autonomous Agents

- Difference Between **LLM Applications** And Agent-Based Systems.
- Autonomous Decision-Making And Goal-Oriented Agents.
- **Single-Agent vs Multi-Agent** Architectures.

Topic 2: Key components of Agentic AI

- Agent Roles, Goals, And Task Decomposition.
- Planning, Reasoning, And Execution Loops.
- **Tool Calling** And External System Integration.
- **Memory Types:** Short-Term, Long-Term, And Episodic Memory.
- Feedback Loops And Self-Reflection Mechanisms.
- Failure Handling And **Recovery Strategies**.

Topic 3: AutoGen

- **Multi-Agent** Conversation Design.
- Role-Based Agent Collaboration.
- Task Delegation And Message Passing.
- **Use Cases** For Autonomous Workflows.

Project 19 AI Data Analyst Agent

Finance

Develop an autonomous agent (e.g. using LangChain or LangGraph) that connects to a database and answers data questions by generating SQL queries and charts.

LLM (GPT-4o/Gemini)

Mistral Agents

Topic 4: CrewAI

- Team-Based **Agent Orchestration**.
- Role Assignment And Hierarchical Agents.
- **Workflow Automation** Using Crews.

Project 20 **Quality Inspector Agent**

Manufacturing

Build agent team to analyze defect images, generate insights, and suggest corrective actions collaboratively.

Vision models

OpenCV tools

CrewAI

Topic 5: LangGraph

- **State-Based Agent** Workflows.
- Directed Graph Execution For Agents.
- Conditional Routing And **Branching Logic**.
- Multi-Step Planning And Execution Graphs.
- **Error Handling** And Retry Mechanisms.

Project 21 **RAG Product Query Bot**

E-Commerce

Build a RAG assistant that answers product comparisons and queries using catalog data and customer reviews.

FAISS / Chroma

OpeAI LLM

LangChain

Topic 6: MCP (Model Context Protocol) / ACP / A2A

- Model Context Protocol (**MCP**) Fundamentals.
- Agent Communication Protocol (**ACP**).
- **Agent-to-Agent** (A2A) Interaction Patterns.
- Context Sharing Across Distributed Agents.
- **Secure Context Exchange** Between Models.
- Enterprise Interoperability Standards For Agents.

Project 22 MCP Coding Assistant & Debugger

IT

Build MCP-powered coding assistant that reads code files, runs tests, debugs errors, and suggests fixes automatically.

MCP SDK

OpenAI LLM

AutoGen

Topic 7: No-Code/Low-Code AI Agent Basics

- Build AI agents using **drag-and-drop workflow** builders.
- **Automate** tasks, approvals, and decision flows without coding.
- Evaluate feasibility, cost, and business **ROI** of AI agents.
- Work confidently with tech teams to scale AI faster.

Project 23 No-Code AI Agent Workflow Builder

Digital Transformation

Create a no-code AI agent workflow that automates business tasks, approvals, and decision flows using visual builders and rule-based orchestration.

LLM API

Rule Engines

n8n / Make.com

Project 24 Interoperable Multi-Agent System using MCP / ACP / A2A

Enterprise AI

Design a distributed multi-agent system where agents communicate, share context, and collaborate securely using standardized protocols across tools and services.

MCP SDK

AutoGen

A2A Protocol

Topic 8: AWS Bedrock

- Foundation Models Available In Bedrock.
- Building Agents Using Bedrock APIs.
- Tool Invocation And Workflow Automation.
- **Security, IAM, And Enterprise Controls.**
- Integrating Bedrock With AWS Services.

Project 25 Autogen Vehicle Diagnostics Agent

Automotive

Create diagnostic agent that interprets sensor logs and suggests repair actions using reasoning and tool execution.

AutoGen

AWS Bedrock

Claude

Project 26 LangGraph Human-in-the-Loop Approval Workflow

Enterprise

Agentic workflow with interrupt nodes requiring human approval before critical actions (e.g. email send, DB write).

LangGraph

FastAPI

Topic 1: LLM Lifecycle, Versioning & Deployment Basics

- End-to-end LLM lifecycle from development to production.
- Model and prompt versioning using evaluation traces.
- API-based LLM deployment using FastAPI and OpenAI / Azure OpenAI endpoints.
- LangSmith, Langfuse, Phoenix, Arize for observability

Project 27 Chatbot Deployment

Tech Support

Build an LLMOps workflow to deploy, version, and monitor a chatbot using automated evaluation and logging.

FastAPI

Docker

LangSmith

Topic 2: Containerization & Inference Serving

- Packaging GenAI applications for consistent deployment.
- Serving LLM-powered APIs using containers.
- Local and cloud-ready inference setups.
- vLLM — PagedAttention for high-throughput open-source model serving
- Text Generation Inference by HuggingFace for production model hosting
- Kubernetes basics for scaling LLM services in cloud environments

Project 28 Production-Grade LLM Application Deployment

IT

Deploy an enterprise-ready LLM application with API endpoints, versioning, monitoring, and controlled rollouts for real-world usage.

FastAPI

Docker

LangSmith

Topic 3: RAG & Agent Deployment Pipelines

- Deploying Retrieval-Augmented Generation pipelines reliably.
- Vector database operations for production workloads.
- Deploying agent workflows with tool calling and state handling.
- **LangGraph** Platform — cloud-hosted agent deployment with persistent state

Project 29 RAG Q&A Pipeline

Knowledge Management

Build a Retrieval-Augmented Generation (RAG) pipeline using a vector database and LLM for Q&A.

Pinecone

HuggingFace

LangChain

RAGAs

Topic 4: Monitoring, Cost Optimization & Reliability

- Tracking latency, token usage, and failures.
- Prompt evaluation and trace-based monitoring.
- Cost optimization, rollback, and safe production controls.
- OpenTelemetry integration for LLM tracing standardization
- Token budget management and prompt compression techniques

Project 30 Agentic AI Monitoring & Cost Optimization System

Deploy an agent-based system that monitors LLM usage, detects performance or cost anomalies, and recommends optimization actions in production environments.

AutoGen

LangGraph

LLM